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Experiments with the Electronic Speed Controller

In this experiment, you conduct experiments with the motor drive system of the McMaster Autonomous Electric Vehicle (AEV). Before you begin the experiments, please read the instructions in this manual carefully. **During the experiments, please follow all the instructions given in this manual.** Please note that the location, size, and models of some of the components shown in the figures in this manual might be slightly different from the components you are using in your project.

Important Safety Procedures to Follow

- Please be careful while you are in the lab and working with the vehicle.
- The experiments in this manual must be conducted in a laboratory environment under the supervision of teaching assistants (TAs). Please follow TA's instructions carefully.
- **Throughout the experiments given in this manual, the wheels of the McMaster AEV should be elevated or removed so that the vehicle does not move.**
- There must be no loose wires or metal pieces on your test bench.
- You must keep terminals and connections out of the way from accidental touching and from contact to test equipment or parts connected to other voltage levels.
- In some experiments, you might need to hold the vehicle to prevent excessive vibrations.
- **You must not touch the vehicle or any components of the vehicle when the motor drive is powered, and the motor is operating.**
- Due to the heat generated in the motor drive system and as the motor consumes battery power, please do not take too much time for each measurement.
- No food or drinks are allowed in the lab.
- **You must always check with the TA before you connect the battery to VESC and before you power up the motor drive.**
- **You must not adjust the motor drive parameters outside of the values provided in this manual.**
- If you encounter any issues with the motor drive operation, you must always check with the TA.
- Stop the operation of the motor first through VESC Tool, then disconnect the VESC Tool from VESC, and then disconnect the battery from VESC.

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Objective 1: Review of the Lab Manual

Before starting your experiments, all members of your group must read and understand the lab manual. Then, you must explain to the TA what needs to be done in the lab, how you will conduct the experiments, and the safety precautions you need to consider. You cannot start the experiments before you answer your TA's questions.

It is mandatory that group members rotate after each objective. Each group member must contribute equally to the experiments. **In your report, you must state which group member lead the experiments in which objective.**

Objective 2: Experiments

In the VESC Tool, under the Experiments tab, you can test the response of the motor as shown in Figure 1. For this objective, only sweep the Speed command. **Maximum speed should be 7500 rpm. Press the run button circled in red.** View and record the process under the Plot tab at the top of the screen.

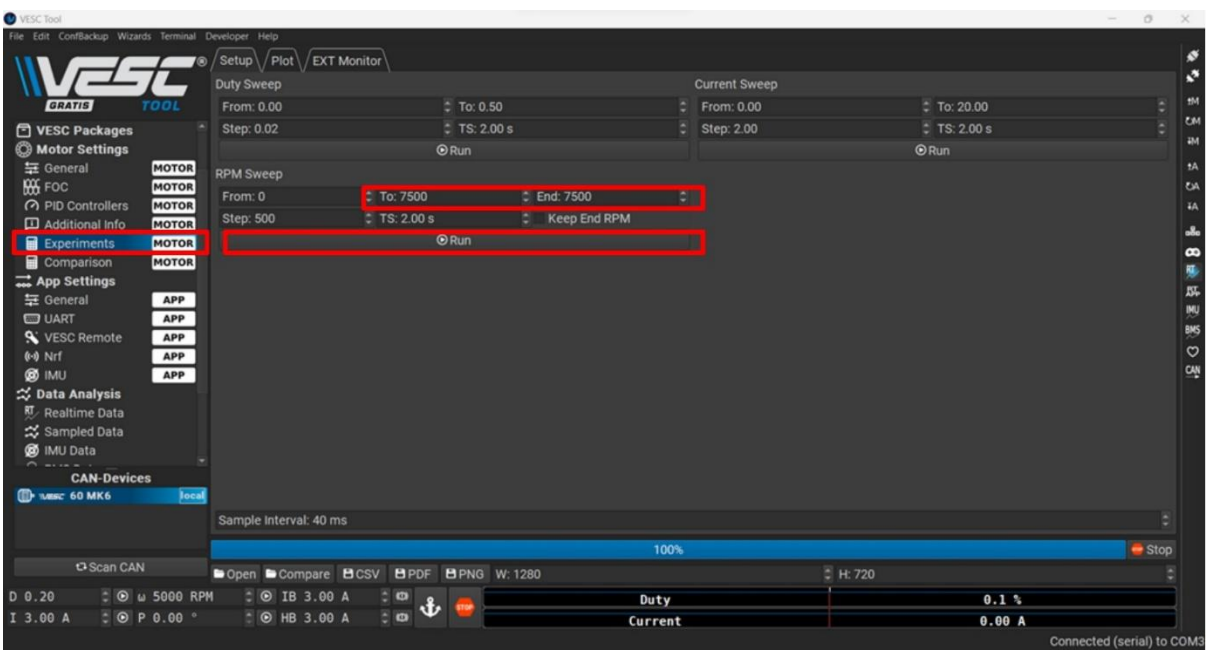

Figure 1: Experiment tab in VESC Tool

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Question 1: Record the experiment results in a .csv file and plot them for your report. Explore how you can log the experiment with Realtime Data feature in the VESC Tool. Attach a screenshot of your plot tab in your report. Attach four graphs from your .csv file: one for RPM, one for q-axis current, one for voltage, and one for duty cycle. Explain how these variables change as the speed increases. What is the relationship between duty cycle, voltage, current, and speed?

Question 2: Conduct a similar experiment and sweep the duty cycle command. **The maximum duty cycle should be 0.2.** Record the speed, q-axis current, voltage, and duty cycle. Plot the result. Compare the results with those from Question 1 when the speed command was swept. What is the difference in speed, current, and voltage response? Explain the reason why.

Question 3: Conduct an experiment where you operate the motor at 0.2 duty cycle. When the motor speed reaches steady state, stop the motor operation using STOP button and record the real-time speed, current, and voltage waveforms. Then conduct the same experiment and, when stopping the motor, apply brake current command (IB) of 1A. Conduct the same experiments multiple times and increase the break current gradually up to 15A. Record the data. How does the speed response change? How does the q-axis current change? Explain the reason in the changes using the operating principles of the motor.

Question 4: Conduct an experiment where you apply a current command. Apply a current command of 8A. **Note that you should not apply more than 8A to the motor at no load condition**. Record the real-time speed, current, duty cycle, and voltage data. When the motor speed is at steady state, reduce the current command to 4A. Record the same data. Note down the speed at steady-state. Explain how the speed, current, voltage, and duty cycle changed.

Stop the motor. Apply a constant current command of 4A. Record the waveforms and explain the operation of the motor. What happened?

Stop the motor. Apply a speed command for the speed you noted down earlier. Record the data and compare the results with those from the previous experiments. How did the motor operation change? Why?

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Objective 3: Record Sampled Data

The VESC Tool has a feature to capture sampled data from the VESC. With this feature, it is possible to plot the phase current waveform instead of the quadrature-axis current magnitude. To do so, click the Sampled Data option under Data Analysis in the left-hand side menu, as shown in Figure 2.

Question 5: Explore the Sampled Data feature in the VESC Tool. For constant speed operation in FOC mode, record the sampled phase current waveforms. Explore the FFT option in the VESC tool and apply it to the phase current measurements. Report the results by applying the speed command and then click on **Sample Now** button. Attach a screenshot of your FFT plot in your report.

Question 6: What is an FFT plot? Why is an FFT plot applied to the phase current measurements important when analyzing the behavior of a motor?

Figure 2: Sampled data feature in VESC Tool

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Question 7: For 2500 rpm, 5000 rpm, 7500 rpm, record the motor phase currents and phase back-emfs using the Sampled Data feature. For each speed and each phase, plot the phase current and phase back-emf on the same graph. You might need to adjust the scale of the current and voltage to show them in the same graph. What is the phase shift between the phase current and phase back-emf at each speed and for each phase? How does the frequency of the phase current and phase back-emf change with speed? Explain the reasons using the electric motor fundamentals.

Appendix: How to modify .csv file into proper columns and make a graph

Step 1: Select column A and then select the Data tab. Select the Text to Columns button.

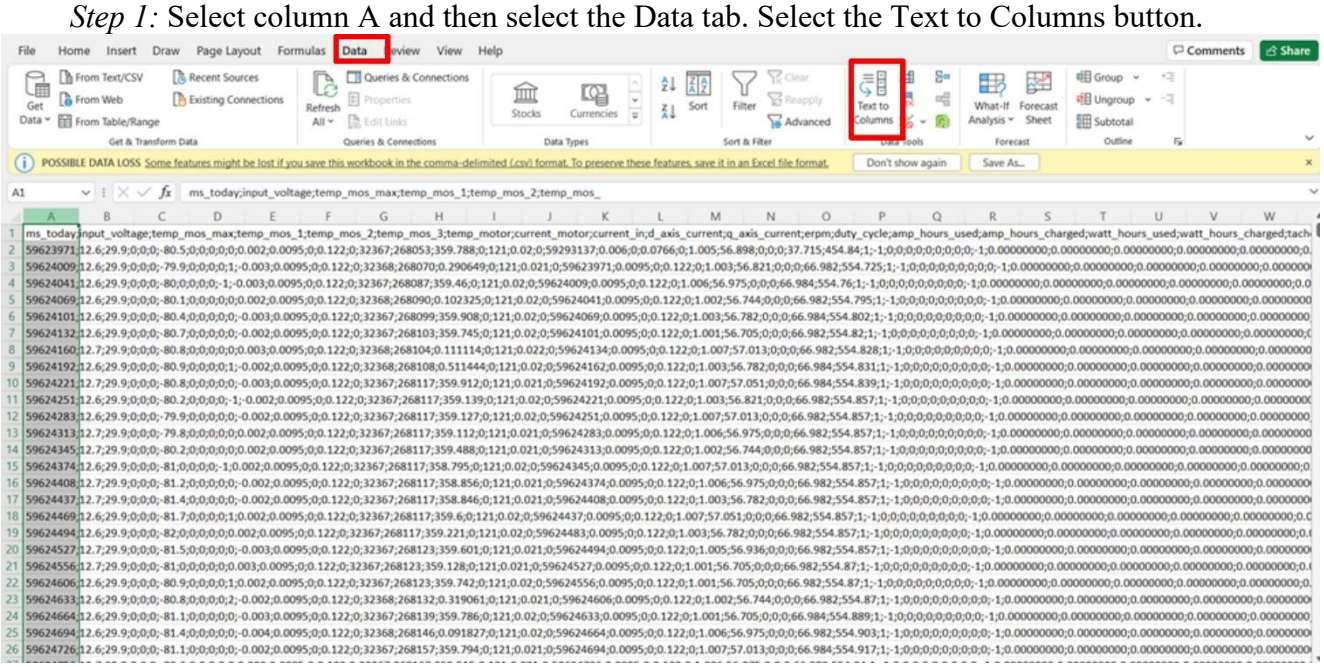

Figure 20: Using the text to columns option in Excel

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Step 2: Select the delimited option

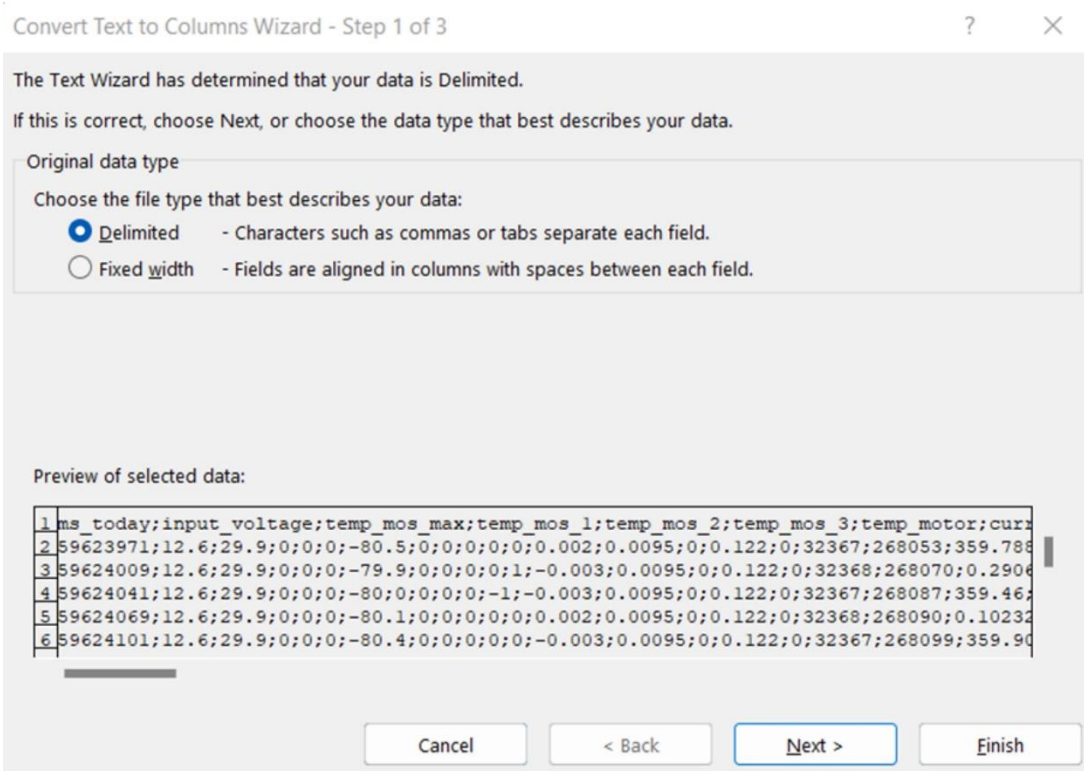

Figure 21: Selecting the delimited option

Step 3: Select the semicolon option ONLY!

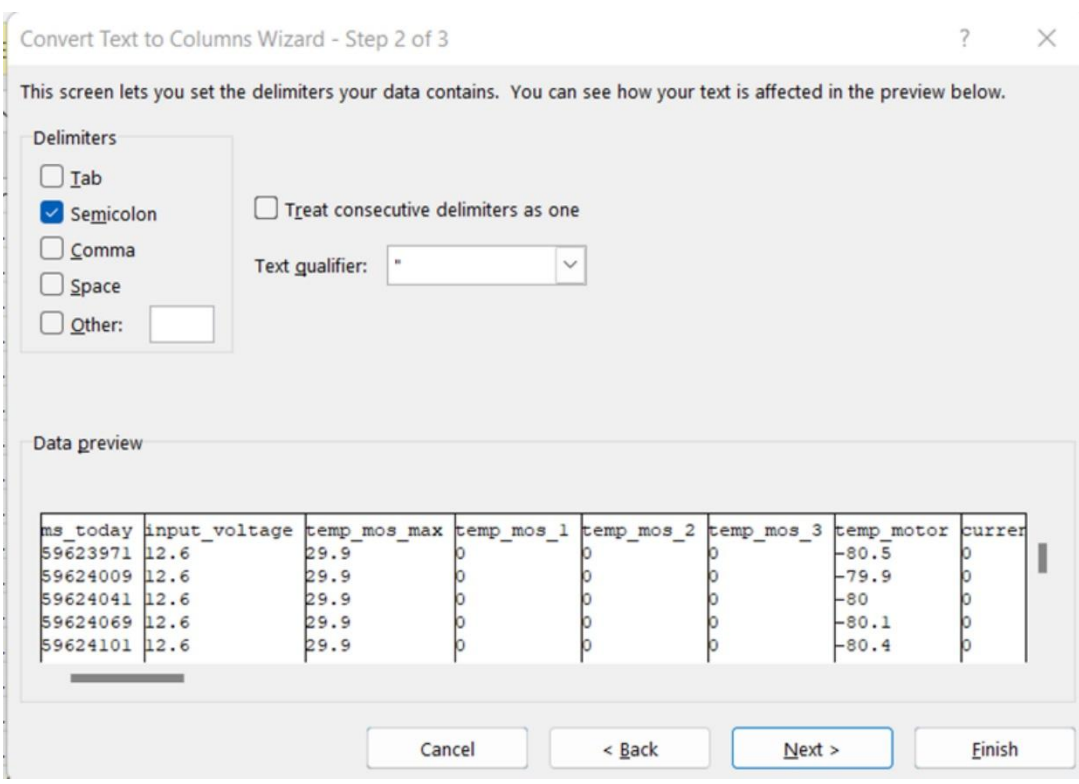

Figure 22: Selecting the semicolon option only

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Step 4: Do not alter anything on this page and select finish.

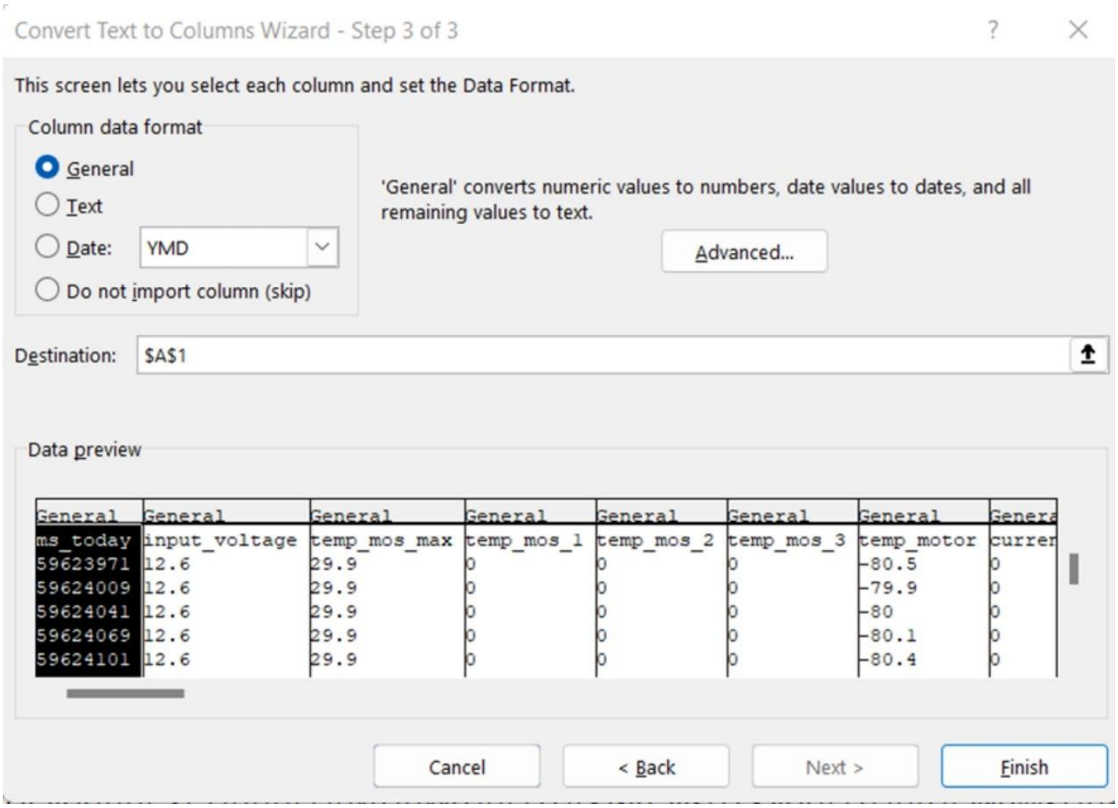

Figure 23: Selecting finish

Step 5: Go to the insert tab and select the columns you need for your graph. Select the graph type from drop down menu highlighted in a red box.

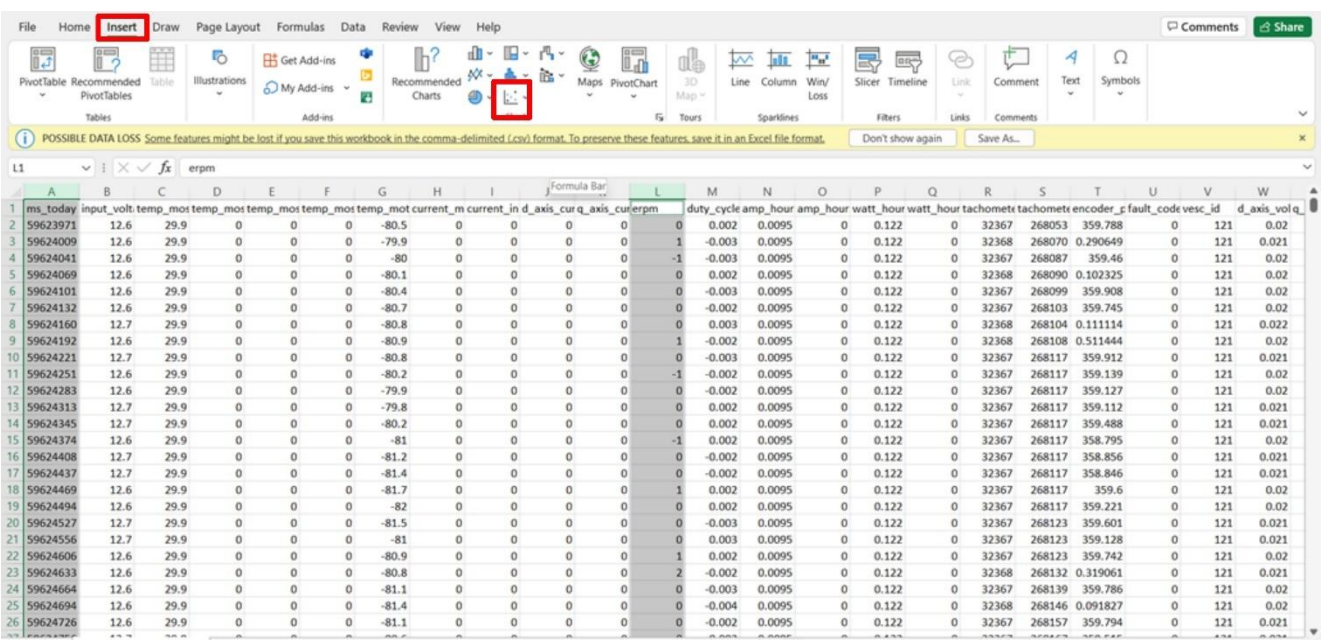

Figure 24: Steps to make a graph in Excel

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Step 6: Select the scatter graph with smooth lines highlighted in a red box.

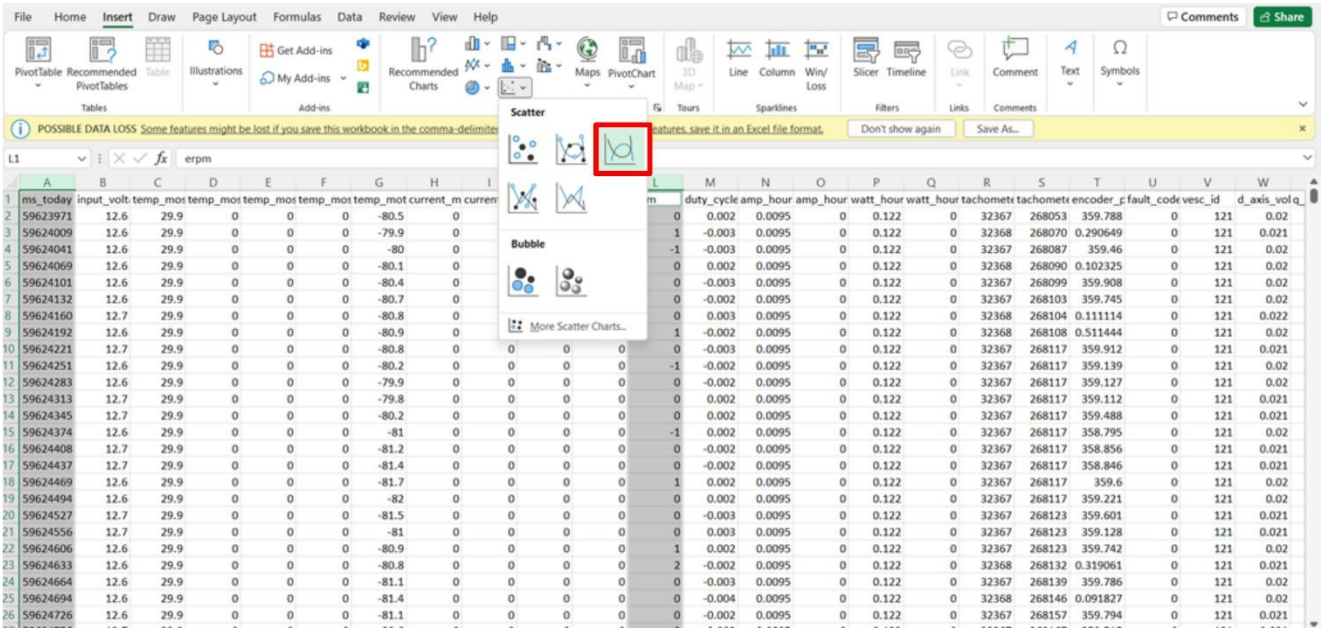

Figure 25: Graph type in Excel

Step 7: Select the plus sign in the highlighted box. Then select axes title and edit the axis titles on the graph to the names you need.

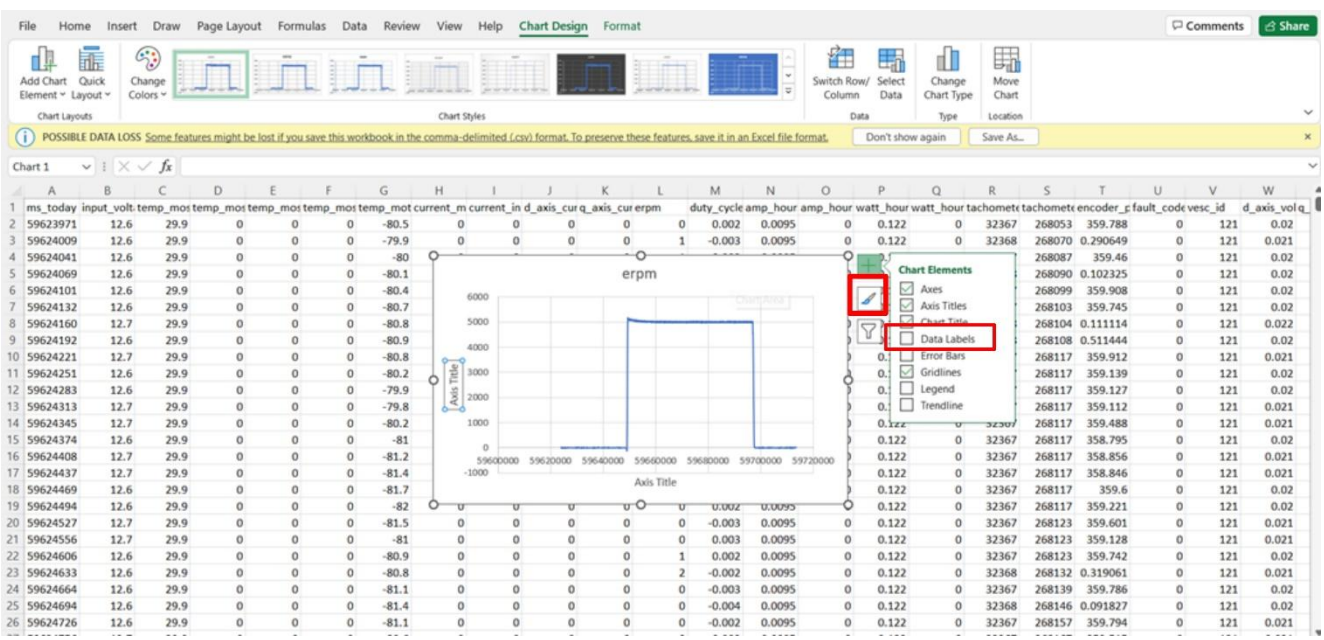

Figure 26: Altering axis titles on a graph